

Notes on Spark Project

Paulinha

Overview

Costs and benefits are inherent aspects of species interactions. Costs associated with mutualistic interactions, such as pollination and seed dispersal, involve production of nectar and fruit from a plant perspective and from an animal perspective there are costs associated with finding and acquiring nectar as well as digesting and processing fruits. The benefits are associated with the provision of resources to the animals and the reproduction to plants. Beyond the importance of the mutualistic interactions to the persistence of interacting species, these interactions also provide ecosystem services, such as production of crops and landscape connectivity. The costs and benefits associated with species interactions are not fixed and are subjected to changes due to environmental conditions, which scale up affecting the provision of ecosystem services.

We developed a framework to explore how costs and benefits associated with species interactions, and consequently ecosystem services, determine the persistence of species and the robustness of service provision under environmental changes. Our modeling framework combines cost-benefit analysis defining fitness functions that determines the outcome of a Boolean process. The Boolean process establishes which species are present in the community given the costs and benefit associated with interactions between species. The presence of species determines whether ecosystem services will be provided. Our model detects transition phases in mutualisms based on increasing costs of interactions associated with environmental changes. We explored how network structure influences species persistence and ecosystem services provision with both theoretical random networks as well as networks parameterized using empirical data for different sampled cost/benefit functions.

Given the broad nature of the modeling framework we took two complementary approaches. Within the first approach we explored how the processes incorporated in the model lead to network structures that resemble empirical ones. The second approach takes empirical networks as a starting point and explores the consequences of the processes incorporated in the model to the persistence of mutualisms and ecosystem services. These two approaches are explored concomitantly and attack (i) how processes lead to patterns and (ii) the consequences of processes in the emerging patterns.

Methods

The two approaches share the same modeling framework's backbone. The common shared aspects of both projects are presented below and afterwards each approach described its own questions and peculiarities. In both cases, a bipartite network of interacting species is defined (either theoretically or empirically). The two sets of nodes can represent plants and pollinators and/or plants and frugivores. Interactions are established between nodes from different sets and each interaction is associated with costs and benefits. The balance between costs and benefits defines which species will be present in the community following a decision function.

Interaction networks

We use our modeling framework as a benchmark to understand what distributions of costs and benefits associated with the interactions leads to networks whose structure resemble empirical networks. Starting from a pool of potential interacting species from two groups (i.e., a bipartite structure of plants and pollinators, for example), we assume all species can potentially interact, given a certain connectance and a random structure to define which species interact. *This is a first approximation, we can also use other models to define potential interactions that take into account realistic degree distributions* Once interactions are established, we define the costs and benefits associated with each interaction for each partner.

Cost and benefit functions

The costs and benefits associated with the interactions between species follow an underlying distribution. We used the beta distribution, which is defined on the interval $[0, 1]$ in terms of two parameters α and β and can take diverse shapes, allowing the exploration of a diverse array of combinations of costs and benefits. Additionally, it provides a limit to the maximum benefit (cost) a species can have. The upper limit is k (the degree of the species, assuming it receives the maximum benefit from all its partners, as benefits are an additive function – see Sec. Decision function). The beta distribution was used previously to model interactions in food web models Stouffer et al. (2005).

Each pairwise interaction is hence associated with an underlying cost and an underlying benefit, which might follow different parameterizations from a beta distribution. The balance between costs and benefits defined by the decision function determines whether a species is able to persist in a community.

Decision Function

Our framework considers a Boolean network determining whether species persist in their environment given the costs and benefits associated with the interactions they establish with mutualistic partners, as well as a contribution of physiological costs associated with the environment. The existence of costs and benefits depends on the presence of an interaction encoded in the adjacency matrix that defines which species interact with whom. The decision function for species i is defined as follows:

$$w_i = \underbrace{\sum_j (B_{ij} - C_{ij})}_{\text{biotic}} - \underbrace{\alpha P_i}_{\text{abiotic}}$$

where B_{ij} is the element ij of the matrix B that encodes the benefits, similarly C_{ij} is the element ij of the matrix C that encodes the costs of the interaction, both sampled from beta distributions. For each pairwise interaction, a net benefit is estimated and summed across all interaction partners, which is the biotic component of the decision function. P_i is the physiological cost associated with the baseline survival of species i and parameter α is the environmental effect, which in combination with the physiological cost compose the abiotic component of the decision function. Whenever $w_i > 0$ species i persists in the environment, whereas whenever $w_i \leq 0$ species i goes extinct.

The Ecosystem-Services Matrix

The matrix describing the ecosystem services is composed of services on the rows and species contributing to the services as its columns. This view of multiple species contributing to an ecosystem service recognizes that services are built through species acting as providers of the service per se and species playing a supporting role in the service (Keyes et al. 2021). The structure of the ecosystem service matrix can be explored from a theoretical point of view as well as using specific case studies. For instance, we can connect the structure

of the empirical network of interactions with the structure of the ecosystem services matrix, where species that play a central role in the matrix are associated with certain services whereas species in the periphery of the network are associated with other services. In this example we further explore how these services differ in their robustness to species' loss or to environmental changes.

Results

Model setting and random expectations

The first part of the projects focuses on the effects of the distribution of costs and benefits on the persistence of species interacting in random bipartite networks, following the Erdős-Renyi (ER) model. The system is initialized with n_i species in one set and n_j species in the other set. Interactions are established between species belonging to different sets, and the probability of an interaction between species from different sets is given by p , which is the expected connectance under the ER-model. Following the Boolean rules, we explored the proportion of surviving species given the difference between the first moment of the distribution of costs and of benefits, i.e., given the difference in the expected values of benefits $E[B]$ and costs $E[C]$. For the beta distribution, this difference (d) is given by:

$$\begin{aligned} d &= E[B] - E[C] \\ E[B] &= \frac{\alpha_b}{\alpha_b + \beta_b} \\ E[C] &= \frac{\alpha_c}{\alpha_c + \beta_c} \end{aligned}$$

When the difference is negative ($d < 0$), expected costs surpass the expected benefits for species interactions, and hence, we expect more species to go extinct. Alternatively, when $d > 0$ the expected benefits are greater than the expected cost, allowing a greater number of species to coexist.

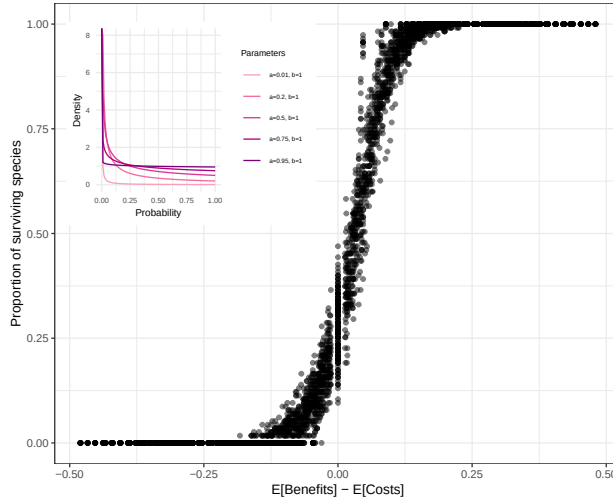


Figure 1: Proportion of surviving species given the difference between the expected value of benefits and costs under beta parameterization considering $\beta = 1$ and $0.01 < \alpha < 0.99$. The inset shows the shape of the distribution for different values of α when $\beta = 1$. The model was run with random networks consisting of 115 nodes (50 on one set of species and 65 on the other set) and expected connectance of $p = 0.65$. Each combination of α_b (associated with benefits) and α_c (associated with costs) had 10 replicates and the proportion of surviving species was recorded.

Figure 1 shows whenever expected costs are greater than expected benefits (negative values on the x -axis), all species go extinct. As the difference decreases, an increasing proportion of species remain in the system. Finally, when expected benefits are much greater than expected costs (positive values on the x -axis), all species remain in the system. This pattern holds for other parameterizations of the beta distribution 2, and alternative choice of distribution (e.g., Lognormal distribution).

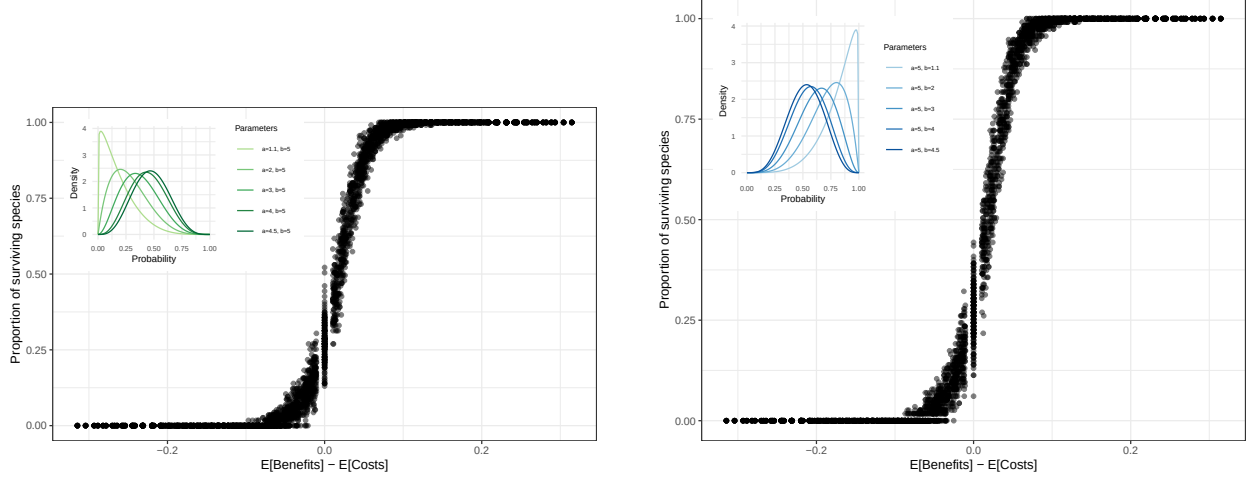


Figure 2: Proportion of surviving species given the difference between expected value of benefits and expected value of costs under beta distributions with parameters following the inset graphs.

When modeling costs and benefits using a Lognormal distribution. In this case, the expected values are given by $E[X] = e^{(\mu + \frac{\sigma^2}{2})}$. For the lognormal distribution, there is a greater variance in the range of parameter values leading to coexistence than observed for the beta distribution. Even with a greater variance, both distributions lead to an S -shaped pattern on the probability of coexistence. These analyses suggest a robust pattern in how asymmetries in the distribution of costs and benefits shape species coexistence.

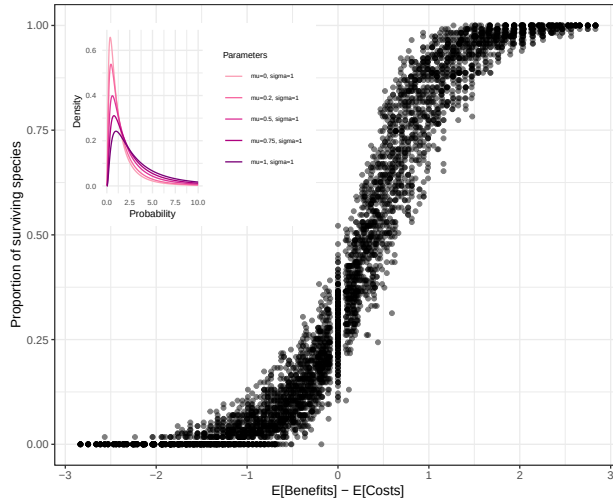


Figure 3: Proportion of surviving species given the difference between expected value of benefits and expected value of costs under lognormal distribution with parameters following the inset graphs. This set of simulations also considered random networks with 115 nodes (50 on one set and 65 on the other set and expected connectance of $p = 0.65$).

Simulations explored parameter combinations in which the final number of surviving species is variable.

Probability of extinction

Under the ER-model degree distribution is homogeneous, with the mean expected degree $\langle k_i \rangle = p(n-1)$ and the variance of the degree distribution $\sigma^2 = p(1-p)(n-1)$. Even if homogeneous, the distribution has a variance and the probability of extinction of a species could be related to their initial degree. Nevertheless, for networks that follow the ER-model, there was no relationship between initial degree and probability of extinction, regardless of specific choice of parameters associated with costs and benefits. Figure 4 shows a series of histograms with the degree distribution resulting from 100 simulations for each panel with a combination of parameters for the distribution of costs and benefits. Darker colors represent the initial degree of surviving species whereas lighter colors represent the initial degree of species that went extinct during the simulation. Distributions are overlapping showing there's no effect of initial degree on species' extinction.

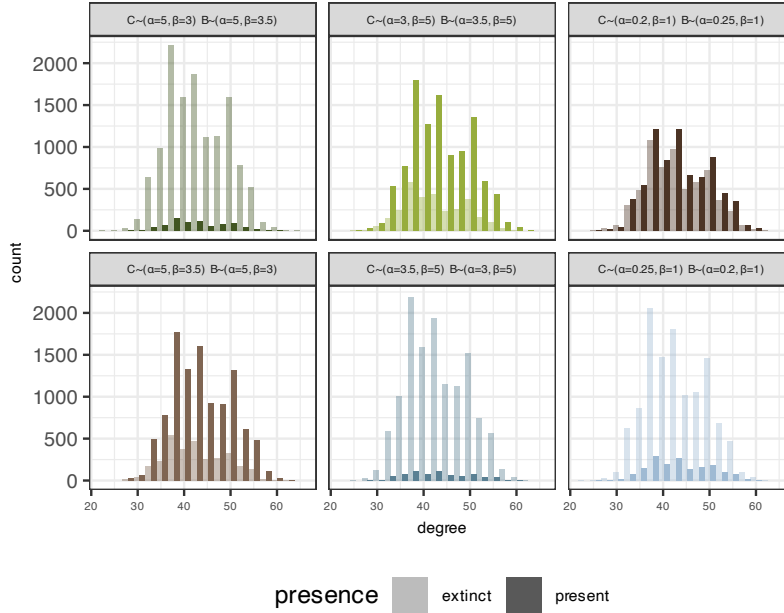


Figure 4: Initial degree distribution at the beginning of the simulation, accounting for species that survived (darker colors) and species that went extinct (light colors). These sets of simulations considered random networks with 115 nodes (50 on one set and 65 on the other set and expected connectance of $p = 0.65$) with 100 replicates for each parameter combination. Each panel presents a different combination of parameters for the distribution of costs (C) and benefits (B).

Effect of connectance

Connectance (number of realized interactions given all possible interactions) can influence dynamics, from the time it takes for dynamics to stabilize to the final number of surviving species. The number of time steps it takes for species' number to stabilize does not depend on network connectance nor on specific parameterization of the beta distribution describing costs and benefits. The mean time across all simulations is 3.75 with a standard deviation ranging from $sd = 1.66 - 2.52$.

The final proportion of surviving species, however, is sensitive to both parameter combination underlying interactions and the connectance of the network. When costs are greater than benefits (panels, 1, 5 and 6 in Figure 6) increasing connectance decreases the final number (proportion) of surviving species. When benefits are slightly higher than costs (panels 2,3, and 4 in Figure 6) increasing connectance in the system favors a greater proportion of surviving species.

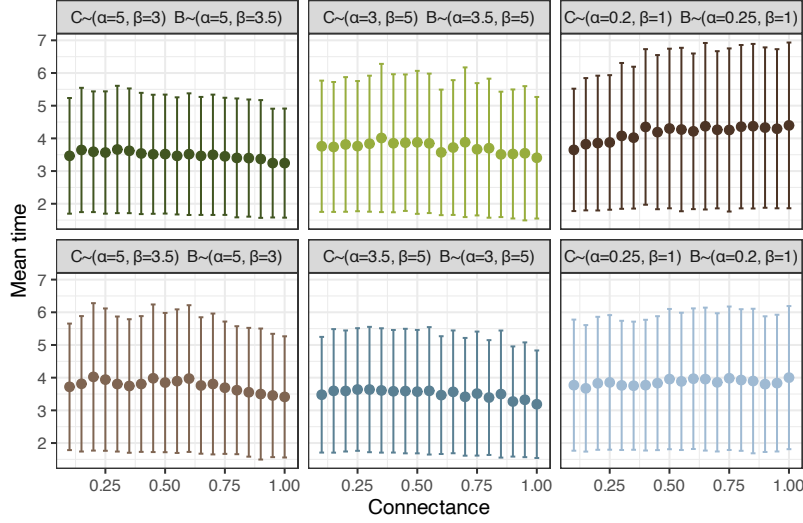


Figure 5: Mean (and standard deviation) of the number of time steps until dynamics stabilize, i.e. no changes in the number of species in the community, as a function of network connectance.

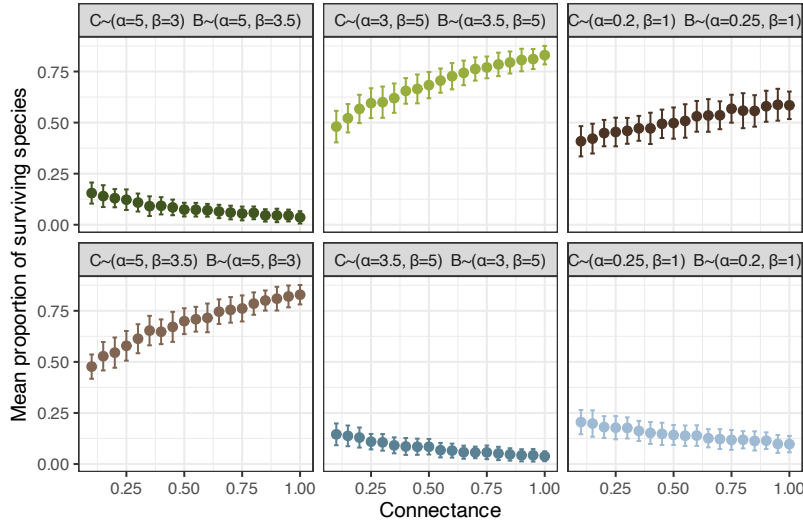


Figure 6: Proportion of surviving species as a function of network connectance.

Follow-up questions

How is the robustness of food webs related to the ecosystem services they provide? (Keyes et al. 2021) explored the consequences of secondary extinctions to the maintenance of services in a food web system. They found that food web and service robustness are highly correlated but service robustness is dependent on trophic level. Additionally, they also found that species providing services **do not** necessarily play a critical role in stabilizing food webs and that species that play supporting roles in service are the critical ones to robustness (of food webs and services). How dependent are these results to the underlying system providing the service? In their case, services are provided through a food web, should we expect different results for pollination or seed dispersal systems? In other words *how does the nature of the underlying ecological system influences the robustness of ecosystem services they provide?*

Supplementary Material

Boundary conditions for costs and benefits and the persistence/extinction of species

What are the boundary conditions for the expectation of having all species present (*benefits* > *costs*) and for having all species extinct of the system (i.e., when *benefits* < *costs*)? In other words, we want to know what is the probability of having the benefits surpass the costs, resulting in all species present in the network, and alternatively, what is the probability of having all species removed from the network if the costs of interaction surpass their benefits. The solution to this problem, considering costs and benefits distributed following independent beta distributions, relies on Appell F1 Hypergeometric function and was derived by Pham-Gia, Turkkan, and Eng (1993). Since the Boolean function determining whether a species will be present is given by a positive fitness, this can be achieved by a species having greater benefits than costs for its interactions (i.e. $\theta_b > \theta_c$), with $\theta_b = X_1 \sim \text{Beta}(\alpha_1, \beta_1)$ and $\theta_c = X_2 \sim \text{Beta}(\alpha_2, \beta_2)$. We want to determine the distribution of the difference $\theta_d = \theta_b - \theta_c$. This difference is piece wise and given by ((Pham-Gia, Turkkan, and Eng 1993)):

Considering $A = \text{Beta}(\alpha_1, \beta_1) \cdot \text{Beta}(\alpha_2, \beta_2)$ and $\cdot$ a product.

For $-1 \leq \theta_d < 0$, meaning costs are greater than benefits, we have:

$$p(\theta_d) = \text{Beta}(\alpha_2, \beta_1) \theta_d^{\beta_1 + \beta_2 - 1} (1 - \theta_d)^{\alpha_2 + \beta_1 - 1} \cdot F_1(\beta_1, \alpha_1 + \beta_1 + \alpha_2 + \beta_2 - 2, 1 - \alpha_1; \beta_1 + \alpha_2; 1 - \theta_d, 1 - \theta_d^2) / A$$

For $0 < \theta_d \leq 1$, with benefits greater than costs, we have:

$$p(\theta_d) = \text{Beta}(\alpha_1, \beta_2) \theta_d^{\beta_1 + \beta_2 - 1} (1 - \theta_d)^{\alpha_1 + \beta_2 - 1} \cdot F_1(\beta_2, 1 - \alpha_2, \alpha_1 + \beta_1 + \alpha_2 + \beta_2 - 2; \alpha_1 + \beta_2; 1 - \theta_d^2, 1 + \theta_d) / A$$

And finally, when the difference between the distribution of benefits and costs is zero $\theta_d = 0$ with $\alpha_1 + \alpha_2 > 1$ and $\beta_1 + \beta_2 > 1$ we have:

$$p(\theta_d) = \frac{\text{Beta}(\alpha_1 + \alpha_2 + 2 - 1, \beta_1 + \beta_2 - 1)}{A}$$

F_1 is the Appell F_1 Hypergeometric Function with the following form:

$$F_1(a, b_1, b_2; c; x, y) = \sum_{m=n=0}^{\infty} \frac{(a)_{m+n} (b_1)_m (b_2)_n}{(c)_{m+n} m! n!} x^m y^n$$

with $(q)_m$ representing the Pochhammer symbol, indicating the rising factor. Substituting the parameters, for the negative case ($-1 \leq \theta_d < 0$) the parameters are $a = \beta_1$, $b_1 = \alpha_1 + \beta_1 + \alpha_2 + \beta_2 - 2$, $b_2 = 1 - \alpha_1$; $c = \beta_1 + \alpha_2$; $x = 1 - \theta_d$, $y = 1 - \theta_d^2$. For the positive case ($0 < \theta_d \leq 1$) the parameters are $a = \beta_2$, $b_1 = 1 - \alpha_2$, $b_2 = \alpha_1 + \beta_1 + \alpha_2 + \beta_2 - 2$; $c = \alpha_1 + \beta_2$; $x = 1 - \theta_d^2$, $y = 1 + \theta_d$.

Other potential distributions

The choice of the beta distribution is supported by biological arguments but it is still an arbitrary choice. Other possible distributions of costs and benefits include:

Lognormal distribution As costs and benefits should, in principle, have the same sign, it is more appropriate to use a distribution defined for positive values. The derivation for the lognormal distribution is not as straight forward as the one for the normal distribution, given we are interested in the difference between two i.i.d. random variables drawn from a lognormal distribution. One solution to find the probability that all species are present is through numerical integration assuming both values are drawn from $\ln N(0, 1)$.

Alternatively, (Parham 2023) proposes a solution describing the Difference-of-Log-Normals (DLN), characterizing its PDF, CDF, moments and parameter estimators.

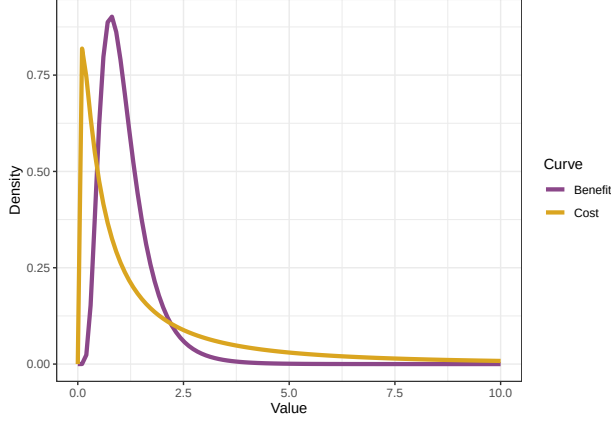


Figure 7: Costs and benefits drawn from a lognormal distribution

Normal distribution Assuming normally and independently distributed costs $C \sim N(\mu_c, \sigma_c^2)$ and benefits $B \sim N(\mu_b, \sigma_b^2)$, we want first to know $P(B > C)$.

$$\begin{aligned} P(B > C) &= P(B - C > 0) \\ &= 1 - P(B - C \leq 0) \end{aligned}$$

If B and C are independent and normally distributed so is their difference, with the expectation of their mean as $\mathbb{E}(B - C) = \mathbb{E}(B) - \mathbb{E}(C) = \mu_b - \mu_c = \mu$ and of their variance as $\text{Var}(B - C) = \sigma_b^2 + \sigma_c^2 = \sigma$. Now we subtract the mean and divide by the variance so that our difference is normally distributed with $\mu = 0$ and $\sigma^2 = 1$ such that $\frac{B-C-\mu}{\sigma} \sim N(0, 1)$. And so:

$$P(B > C) = 1 - \Phi\left(\frac{-\mu}{\sigma}\right) \quad (1)$$

where Φ is the cumulative distribution function of $N(0, 1)$.

Now let's consider an example: if species benefits have the distribution $B \sim N(1.2, 2)$ and costs are distributed as $C \sim N(2, 1.5)$, the distribution describing these values is:

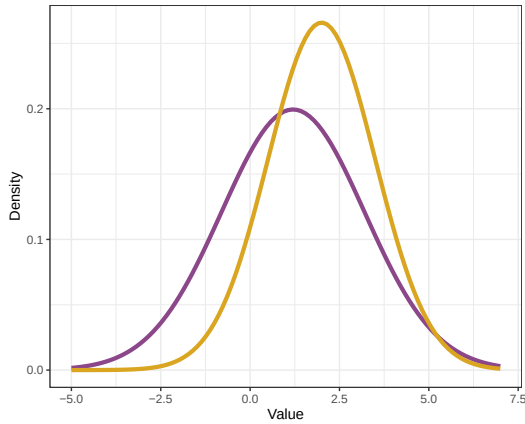


Figure 8: Example of cost-benefit functions following a normal distribution

Using the derived expression, the resulting probability of having all species present in this system, i.e., the $P(B > C)$, meaning benefits are greater than costs is: 0.3344646. Alternatively, the probability that all

species goes extinct in this example is: 0.6655354.

In more general terms, we can explore how the difference between benefits and costs influences the probability of all species coexisting. Assuming a standard deviation of $\sigma^2 = 1$ we explore the effect of the mean difference between costs and benefits in species' probability of persistence, deriving the probability according to equation 1.

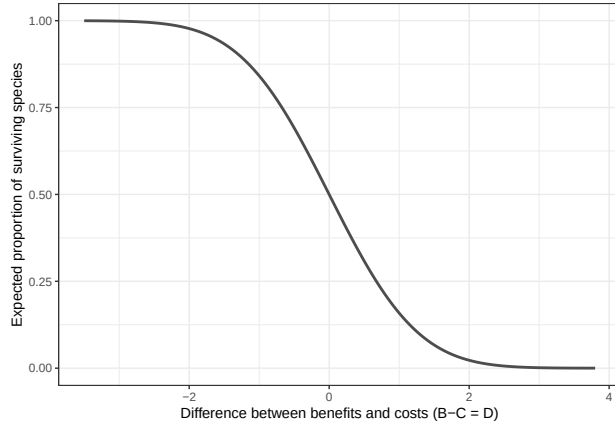


Figure 9: Expected proportion of species' persistence given the difference in benefits and costs

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